

A.4.3. MULTIPLE KERNEL LEARNING (PROPOSITION 8)

Multiple kernel learning (Gönen & Alpaydm, 2011) also falls into our setting. A special case of this family of kernels was studied for the (easier to analyze) “streaming” MMD estimator by Gretton et al. (2012b).

(V) Let $\{k_i\}_{i=1}^D$ be a set of base kernels, each satisfying $\sup_{x \in \mathcal{X}} k_i(x, x) \leq K$ for some finite K . Define k_ω as

$$k_\omega(x, y) = \sum_{i=1}^D \omega_i k_i(x, y).$$

Define the norm of a kernel parameter by the norm of the corresponding vector $\omega \in \mathbb{R}^D$. Let Ω be a set of possible parameters such that for each $\omega \in \Omega$, k_ω is positive semi-definite, and $\|\omega\| \leq R_\Omega$ for some $R_\Omega < \infty$.

Not only does learning in this setting work (Proposition 26), it is also – unlike the deep setting – efficient to find an exact maximizer of $\hat{J}_\lambda$ (Proposition 27).

Proposition 26. *Assumption (V) implies Assumptions (A) to (C). In particular,*

$$\begin{aligned} \sup_{\omega \in \Omega} \sup_{x \in \mathcal{X}} k_\omega(x, x) &\leq K R_\Omega \sqrt{D} \\ |k_\omega(x, y) - k_{\omega'}(x, y)| &\leq K \sqrt{D} \|\omega - \omega'\|. \end{aligned}$$

Proof. Assumption (B) is immediate from Assumption (V), since $\Omega \subset \mathbb{R}^D$. Let $\mathbf{k}(x, y) \in \mathbb{R}^D$ denote the vector whose i th entry is $k_i(x, y)$, so that $k_\omega(x, y) = \omega^\top \mathbf{k}(x, y)$. As $\|\mathbf{k}(x, y)\|_\infty \leq K$, we know $\|\mathbf{k}(x, y)\| \leq K \sqrt{D}$. Assumptions (A) and (C) follow by Cauchy-Schwartz. $\square$

Proposition 27. *Take Assumption (V), and additionally assume that $\Omega = \{\omega \mid \forall i. \omega_i \geq 0, \sum_i \omega_i = Q\}$ for some $Q < \infty$. A maximizer of $\hat{J}_\lambda(\omega)$ can then be found by scaling the solution to a convex quadratic program,*

$$\tilde{\omega} = \arg \min_{\omega \in [0, \infty)^D : \omega^\top \mathbf{b} = 1} \omega^\top (\mathbf{A} + \lambda I) \omega, \quad \hat{\omega} = \frac{Q}{\sum_i \tilde{\omega}_i} \tilde{\omega} \in \arg \max_{\omega \in \Omega} \hat{J}_\lambda(\omega),$$

where

$$\begin{aligned} (\mathbf{H}_{ij})_\ell &= k_\ell(X_i, X_j) + k_\ell(Y_i, Y_j) - k_\ell(X_i, Y_j) - k_\ell(X_j, Y_i) \\ \mathbf{b} &= \frac{1}{n(n-1)} \sum_{i \neq j} \mathbf{H}_{ij} \in \mathbb{R}^D \\ \mathbf{A} &= \frac{4}{n^3} \sum_i \left(\sum_j \mathbf{H}_{ij} \right) \left(\sum_j \mathbf{H}_{ij} \right)^\top - \frac{4}{n^4} \left(\sum_{ij} \mathbf{H}_{ij} \right) \left(\sum_{ij} \mathbf{H}_{ij} \right)^\top \in \mathbb{R}^{D \times D}, \end{aligned}$$

as long as $\mathbf{b}$ has at least one positive entry.

Proof. The H matrix used by $\hat{\eta}_\omega$ and $\hat{\sigma}_\omega$ takes a simple form:

$$H_{ij}^{(\omega)} = k_\omega(X_i, X_j) + k_\omega(Y_i, Y_j) - k_\omega(X_i, Y_j) - k_\omega(X_j, Y_i) = \omega^\top \mathbf{H}_{ij}.$$

Thus

$$\begin{aligned} \hat{\eta}_\omega &= \omega^\top \left(\frac{1}{n(n-1)} \sum_{i \neq j} \mathbf{H}_{ij} \right) = \omega^\top \mathbf{b} \\ \hat{\sigma}_\omega^2 &= \frac{4}{n^3} \sum_i \left(\omega^\top \sum_j \mathbf{H}_{ij} \right)^2 - \frac{4}{n^4} \left(\omega^\top \sum_{ij} \mathbf{H}_{ij} \right)^2 \\ &= \omega^\top \left(\frac{4}{n^3} \sum_i \left(\sum_j \mathbf{H}_{ij} \right) \left(\sum_j \mathbf{H}_{ij} \right)^\top - \frac{4}{n^4} \left(\sum_{ij} \mathbf{H}_{ij} \right) \left(\sum_{ij} \mathbf{H}_{ij} \right)^\top \right) \omega = \omega^\top \mathbf{A} \omega. \end{aligned}$$

Note that because $\hat{\sigma}_\omega^2 \geq 0$ for any ω , we have $\mathbf{A} \succeq 0$. We have now obtained a problem equivalent to the one in Section 4 of [Gretton et al. \(2012b\)](#); the argument proceeds as there. $\square$

A.5. Miscellaneous Proofs

The following lemma was used for Propositions 21 and 23.

Lemma 20. *The Gaussian kernel $\kappa(a, b) = \exp\left(-\frac{\|a-b\|^2}{2\sigma^2}\right)$ satisfies*

$$|\kappa(a, b) - \kappa(a', b')| \leq \frac{1}{\sigma\sqrt{e}} (\|a - b\| + \|a' - b'\|) \leq \frac{1}{\sigma\sqrt{e}} (\|a - a'\| + \|b - b'\|).$$

Proof. We have that

$$\begin{aligned} |\kappa(a, b) - \kappa(a', b')| &= \left| \exp\left(-\frac{\|a-b\|^2}{2\sigma^2}\right) - \exp\left(-\frac{\|a'-b'\|^2}{2\sigma^2}\right) \right| \\ &\leq \|x \mapsto \exp\left(-\frac{x^2}{2\sigma^2}\right)\|_L \|a - b\| - \|a' - b'\|. \end{aligned}$$

We can bound the Lipschitz constant as its maximal derivative norm,

$$\sup_x \frac{|x|}{\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right).$$

Noting that

$$\frac{d}{dx} \log\left(\frac{|x|}{\sigma^2} \exp\left(-\frac{x^2}{2\sigma^2}\right)\right) = \frac{1}{x} - \frac{x}{\sigma^2}$$

vanishes only at $x = \pm\sigma$, the supremum is achieved by using that value, giving

$$\|x \mapsto \exp\left(-\frac{x^2}{2\sigma^2}\right)\|_L = \frac{1}{\sigma\sqrt{e}}.$$

The result follows from

$$\| \|a - b\| - \|a' - b'\| \| \leq \|a - b - a' + b'\| \leq \|a - a'\| + \|b - b'\|. \quad \square$$

This next lemma was used in Proposition 23.

Lemma 22. *Under Assumption (II), suppose ω, ω' have $\|\omega\| \leq R, \|\omega'\| \leq R$, with $R \neq 1$. Then, for any x ,*

$$\|\phi_\omega(x)\| \leq R^\Lambda \|x\| + \frac{R}{R-1} (R^\Lambda - 1) \quad (13)$$

$$\|\phi_\omega(x) - \phi_{\omega'}(x)\| \leq \left(\Lambda R^{\Lambda-1} \left(\|x\| + \frac{R}{R-1} \right) - \frac{R^\Lambda - 1}{(R-1)^2} \right) \|\omega - \omega'\|. \quad (14)$$

If $R \geq 2$, we furthermore have

$$\|\phi_\omega(x)\| \leq R^\Lambda (\|x\| + 2) \quad (15)$$

$$\|\phi_\omega(x) - \phi_{\omega'}(x)\| \leq \Lambda R^{\Lambda-1} (\|x\| + 2) \|\omega - \omega'\|. \quad (16)$$

Proof. First, $\|\phi_\omega^{(0)}(x)\| = \|x\|$, showing (13) when $\Lambda = 0$. In general,

$$\begin{aligned} \|\phi_\omega^{(\ell)}(x)\| &= \|\sigma^{(\ell)} \left(W_\omega^{(\ell)} \phi_\omega^{(\ell-1)}(x) + b_\omega^{(\ell)} \right)\| \\ &\leq \|W_\omega^{(\ell)} \phi_\omega^{(\ell-1)}(x) + b_\omega^{(\ell)}\| \\ &\leq \|W_\omega^{(\ell)}\| \|\phi_\omega^{(\ell-1)}(x)\| + \|b_\omega^{(\ell)}\| \\ &\leq R \|\phi_\omega^{(\ell-1)}(x)\| + R, \end{aligned}$$

and expanding this recursion gives

$$\|\phi_\omega^{(\ell)}(x)\| \leq R^\ell \|x\| + \sum_{m=1}^{\ell} R^m = R^\ell \|x\| + \frac{R}{R-1}(R^\ell - 1).$$

Now, we have (14) for $\Lambda = 0$ because $\phi_\omega^{(0)}(x) - \phi_{\omega'}^{(0)}(x) = 0$. For $\ell \geq 1$, we have

$$\begin{aligned} \|\phi_\omega^{(\ell)}(x) - \phi_{\omega'}^{(\ell)}(x)\| &= \|\sigma^{(\ell)} \left(W_\omega^{(\ell)} \phi_\omega^{(\ell-1)}(x) + b_\omega^{(\ell)} \right) - \sigma^{(\ell)} \left(W_{\omega'}^{(\ell)} \phi_{\omega'}^{(\ell-1)}(x) + b_{\omega'}^{(\ell)} \right)\| \\ &\leq \|W_\omega^{(\ell)} \phi_\omega^{(\ell-1)}(x) - W_{\omega'}^{(\ell)} \phi_\omega^{(\ell-1)}(x)\| + \|W_{\omega'}^{(\ell)} \phi_\omega^{(\ell-1)}(x) - W_{\omega'}^{(\ell)} \phi_{\omega'}^{(\ell-1)}(x)\| + \|b_\omega^{(\ell)} - b_{\omega'}^{(\ell)}\| \\ &\leq \|W_\omega^{(\ell)} - W_{\omega'}^{(\ell)}\| \|\phi_\omega^{(\ell-1)}(x)\| + \|W_{\omega'}^{(\ell)}\| \|\phi_\omega^{(\ell-1)}(x) - \phi_{\omega'}^{(\ell-1)}(x)\| + \|\omega - \omega'\| \\ &\leq \|\omega - \omega'\| \left(R^{\ell-1} \|x\| + \frac{R}{R-1}(R^{\ell-1} - 1) + 1 \right) + R \|\phi_\omega^{(\ell-1)}(x) - \phi_{\omega'}^{(\ell-1)}(x)\|. \end{aligned}$$

Expanding the recursion yields

$$\begin{aligned} \|\phi_\omega^{(\ell)}(x) - \phi_{\omega'}^{(\ell)}(x)\| &\leq \sum_{m=0}^{\ell-1} R^m \left(R^{\ell-1-m} \|x\| + \frac{R}{R-1}(R^{\ell-m-1} - 1) + 1 \right) \|\omega - \omega'\| \\ &= \sum_{m=0}^{\ell-1} \left(R^{\ell-1} \|x\| + \frac{R^\ell}{R-1} - \frac{R^{m+1}}{R-1} + R^m \right) \|\omega - \omega'\| \\ &= \left(\ell R^{\ell-1} \|x\| + \frac{\ell R^\ell}{R-1} - \left(\frac{R}{R-1} - 1 \right) \sum_{m=0}^{\ell-1} R^m \right) \|\omega - \omega'\| \\ &= \left(\ell R^{\ell-1} \left(\|x\| + \frac{R}{R-1} \right) - \frac{1}{R-1} \frac{R^\ell - 1}{R-1} \right) \|\omega - \omega'\|. \end{aligned}$$

When $R \geq 2$, we have that $R/(R-1) \leq 2$ and $R^\ell > 1$, giving (15) and (16). $\square$

B. Experimental Details

B.1. Details of synthetic datasets

Table 6 shows details of four synthetic datasets. *Blob* datasets are often used to validate two-sample test methods (Gretton et al., 2012b; Jitkrittum et al., 2016; Sutherland et al., 2017), although we rotate each blob to show the benefits of non-homogeneous kernels. *HDGM* datasets are first proposed in this paper. *HDGM-D* can be regarded as *high-dimension Blob-D* which contains two modes with the same variance and different covariance.

Table 6. Specifications of $\mathbb{P}$ and $\mathbb{Q}$ of synthetic datasets. $\mu_1^b = [0, 0]$, $\mu_2^b = [0, 1]$, $\mu_3^b = [0, 2]$, $\dots$, $\mu_8^b = [2, 1]$, $\mu_9^b = [2, 2]$ (same with Figure 1a). $\mu_1^h = \mathbf{0}_d$, $\mu_2^h = 0.5 \times \mathbf{1}_d$, I_d is an identity matrix with size d . $\Delta_i^b = -0.02 - 0.002 \times (i - 1)$ if $i < 5$ and $\Delta_i^b = 0.02 + 0.002 \times (i - 6)$ if $i > 5$. if $i = 5$, $\Delta_i^b = 0$ (same with Figure 1a). Δ_1^h and Δ_2^h are set to 0.5 and -0.5 , respectively.

Datasets	$\mathbb{P}$	$\mathbb{Q}$
<i>Blob-S</i>	$\sum_{i=1}^9 \frac{1}{9} \mathcal{N}(\mu_i^b, 0.03 \times I_2)$	$\sum_{i=1}^9 \frac{1}{9} \mathcal{N}(\mu_i^b, 0.03 \times I_2)$
<i>Blob-D</i>	$\sum_{i=1}^9 \frac{1}{9} \mathcal{N}(\mu_i^b, 0.03 \times I_2)$	$\sum_{i=1}^9 \frac{1}{9} \mathcal{N} \left(\mu_i^b, \begin{bmatrix} 0.03 & \Delta_i^b \\ \Delta_i^b & 0.03 \end{bmatrix} \right)$
<i>HDGM-S</i>	$\sum_{i=1}^2 \frac{1}{2} \mathcal{N}(\mu_i^h, I_d)$	$\sum_{i=1}^2 \frac{1}{2} \mathcal{N}(\mu_i^h, I_d)$
<i>HDGM-D</i>	$\sum_{i=1}^2 \frac{1}{2} \mathcal{N}(\mu_i^h, I_d)$	$\sum_{i=1}^2 \frac{1}{2} \mathcal{N} \left(\mu_i^h, \begin{bmatrix} 1 & \Delta_i^h & \mathbf{0}_{d-2} \\ \Delta_i^h & 1 & \mathbf{0}_{d-2} \\ \mathbf{0}_{d-2}^T & \mathbf{0}_{d-2}^T & I_{d-2} \end{bmatrix} \right)$